

Genetic and Epidemiologic Characterization of HIV-1 Infection In Liaoning Province, China

Xiaoxu Han, MD,* Di Dai, MS,* Bin Zhao, MS,* Jing Liu, MS,* Haibo Ding, BS,* Min Zhang, MD,* Qinghai Hu, BS,* Chunming Lu, BS,† Mark Goldin, BS,‡ Yutaka Takebe, MD, PhD,§ Linqi Zhang, PhD,|| and Hong Shang, MD, PhD*

Background: Although many studies of HIV-1 in China have focused on high-risk injecting drug users and former blood donors in high prevalence regions, little is known about HIV-1 in relatively low prevalence provinces. Here, we compare the epidemiologic and genetic profile of HIV-1 in Liaoning—a low prevalence province—with those identified in China's most severely affected provinces.

Materials and Methods: Two hundred eight HIV-1-positive subjects from all major cities in Liaoning province were recruited between 2000 and 2008. 2.6 kilobase gag-pol sequences were amplified from plasma viral RNA and sequenced directly. The HIV-1 sequences obtained were analyzed using phylogenetic and recombinant approaches.

Results: We have shown that in recent years, although HIV-1 prevalence in Liaoning has remained low, the rate of new infection has increased rapidly, particularly among men who have sex with men and heterosexual risk individuals (together comprising >54% of infected individuals in 2007). Furthermore, phylogenetic analysis has identified all major subtypes/circulating recombinant forms of HIV-1 in Liaoning previously identified in high prevalence provinces. Our study also shows close relationships between HIV-1 subtype/circulating recombinant form and certain risk behaviors. Notably, men who

have sex with men and heterosexual risk individuals harbor most divergent strains of HIV-1 from multiple high-risk groups.

Conclusions: Our study suggested that HIV-1 continues to spread to the general population through sexual contact; Liaoning, therefore, serves as the critical base for the introduction and spread of HIV-1 in northeast China. We believe the transmission patterns suggested herein will help guide public health workers in reducing further spread of HIV-1 within China.

Key Words: HIV type I, injecting drug users, men who have sex with men, China

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INTRODUCTION

The HIV-1/AIDS epidemic in China is at a critical juncture. Although HIV-1 infection has historically been largely confined to high-risk groups such as injecting drug users (IDUs) and former commercial blood and plasma donors (FBDs) in rural areas,^{1,2} the rate of HIV-1 infection has been increasing rapidly in urban settings among men who have sex with men (MSM) and female sex workers. This is a worrisome trend previously seen in other Asian countries.^{3,4}

The current epidemic in China can be broadly classified into 2 major risk groups. The first major risk group arose from injecting drug use and needle sharing in Yunnan province, and along drug-trafficking routes to the neighboring provinces of Guangxi, Guizhou, and Sichuan and to the far northwest province of Xinjiang.³ The second group arose from illegal and unsanitary blood collection and transfusion practices among FBDs in Henan and neighboring provinces.^{1,2} This typically involved the pooling of blood from several donors of the same blood type, followed by the removal of plasma and the reinfusion of cells back into the original donors to reduce anemia.^{1,2} Contamination of these pooled samples with HIV-1, along with needle reuse and unsterilized equipment, greatly enhanced viral transmission. Epidemiologic and molecular genetic studies indicate that both of these 2 epidemic groups stemmed from the initial infection of Dai and Jingpo ethnic minority IDUs in southwest Yunnan province in the late 1980s.^{5–11} IDUs and FBDs represent the first and worst affected populations in the HIV-1 epidemic in China.

Although there exist many HIV-1 epidemiologic and genetic studies of these 2 major groups,^{3,7–16} little is known

From the *Key Laboratory of AIDS Immunology of Ministry of Health, No 1 Hospital of China Medical University, Shenyang, China; †Liaoning Center For Disease Control and Prevention, Shenyang, China; ‡Albert Einstein College of Medicine, Bronx, NY; §Laboratory of Molecular Virology and Epidemiology, AIDS Research Center, National Institute of Infectious Diseases, Toyama, Shinjuku-ku, Tokyo, Japan; and ||Comprehensive AIDS Research Center, Tsinghua University, AIDS Research Center, Institute of Pathogen, Chinese Academy of Medical Sciences, and Peking Union Medical College, Beijing, China.

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The authors X.H., L.Z., and H.S. contributed equally to this work.

Correspondence to: Hong Shang, MD, PhD, Key Laboratory of AIDS Immunology of Ministry of Health, No 1 Hospital of China Medical University, Shenyang 110001, China (e-mail: hongshang100@hotmail.com) or Linqi Zhang, PhD, Comprehensive AIDS Research Center, Tsinghua University, and AIDS Research Center, Chinese Academy of Medical Sciences, Peking Union Medical College, Beijing, People Republic of China (e-mail: lzhang07@hotmail.com).

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about the situation in relatively low prevalence provinces. As such, we aim to study the epidemiologic and genetic features of HIV-1 in a low prevalence region—Liaoning province of northeast China—and compare with those identified in China's most severely affected provinces. We believe that our study will not only help to monitor the transmission patterns of HIV-1 from high to relatively low prevalence areas but also assist public health workers in implementing measures to curb the spread of HIV-1 within China.

MATERIALS AND METHODS

Study Subjects and Specimens

A total of 208 HIV-1-positive subjects from all major cities in Liaoning province were recruited between 2000 and 2008 by the First Affiliated Hospital China Medical University in Shenyang, Liaoning Province, in northeastern China. These included 109 cases from Shenyang, 31 from Huludao, 20 from Dalian, 16 from Dandong, 6 from Fushun, 4 cases from Chaoyang, Panjin, Tieling, and Yingkou, 3 cases from Anshan and Liaoyang, 2 cases from Jinzhou, and 1 case from Benxi and Fuxin (Fig. 1). This study was approved by the institution's ethical committee, and the study participants were interviewed by clinicians to determine their epidemiologic background, using questionnaires designed by the UNAIDS HIV-1 characterization network. This information was then verified with that from the HIV patient management database in the Hospital. Study subjects included 70 individuals with heterosexual risk, 52 FBDs or transfusion recipients, 10 IDUs, 75 male homosexuals, and 1 child born to an HIV-infected mother. Two hundred six of 208 subjects were adults (>15 years old), including 165 males and 41 females. Adult subject age data were obtained for 93.4% males (range: 19–69 years; mean: 36.4 ± 10.1 years) and 92.3% females (range: 18–58 years; mean: 35.1 ± 10.4 years). The 2 subjects <15

years old (4 and 14 years old) were males. All 208 specimens were serologically determined as HIV-1 positive with no evidence of HIV-2 infections. Baseline CD4⁺ T-cell count was obtained for 95.2% male subjects (mean: 356 ± 243 cells/ μ L) and 100% female subjects (mean: 405 ± 243 cells/ μ L), and baseline HIV-1 viral load was obtained for 73% male subjects (mean: $4.68 \pm 0.95 \times 10^3$ Igcopies/mL) and 92.3% female subjects (mean: $4.65 \pm 0.95 \times 10^3$ Igcopies/mL). All patients were treatment-naïve at the point of sample collection.

RNA Purification, Polymerase Chain Reaction Amplification, Sequencing, and Phylogenetic Analysis

HIV-1 RNA was extracted from plasma by a column purification method (QIAamp. Viral RNA Mini Kit; Qiagen, Hilden, Germany) and subjected to reverse transcription-nested polymerase chain reaction (PCR). The nucleotide sequence of the 2.6 kilobase *gag* reverse transcriptase (RT) gene (nucleotide position relative to HXB2 genome: 790–3421) was determined by direct sequencing (ABI PRISM. 3130 Genetic Analyzer, Applied Biosystems, Foster City, CA). The PCR primer-binding sites for the amplicon were based on published sequences of geographical variants and were as highly conserved as possible. Codon alignment of query sequences with various HIV-1 reference subtypes and circulating recombinant form (CRFs) from the HIV database (<http://hiv-web.lanl.gov/>) was performed. Selected HIV-1 subtypes/CRFs of geographical importance were also included in the alignment. Phylogenetic and molecular evolutionary distances were estimated using the neighbor-joining method and the Kimura 2-parameter model with a transition–transversion ratio of 2.0. The reliability of the branching orders was tested by bootstrap analysis of 1000 replicates. Bootscanning, similarity plots and informative site analysis were performed using SimPlot v3.5 to define the recombination structure. All RNA extractions and PCR amplifications were carried out with appropriate negative

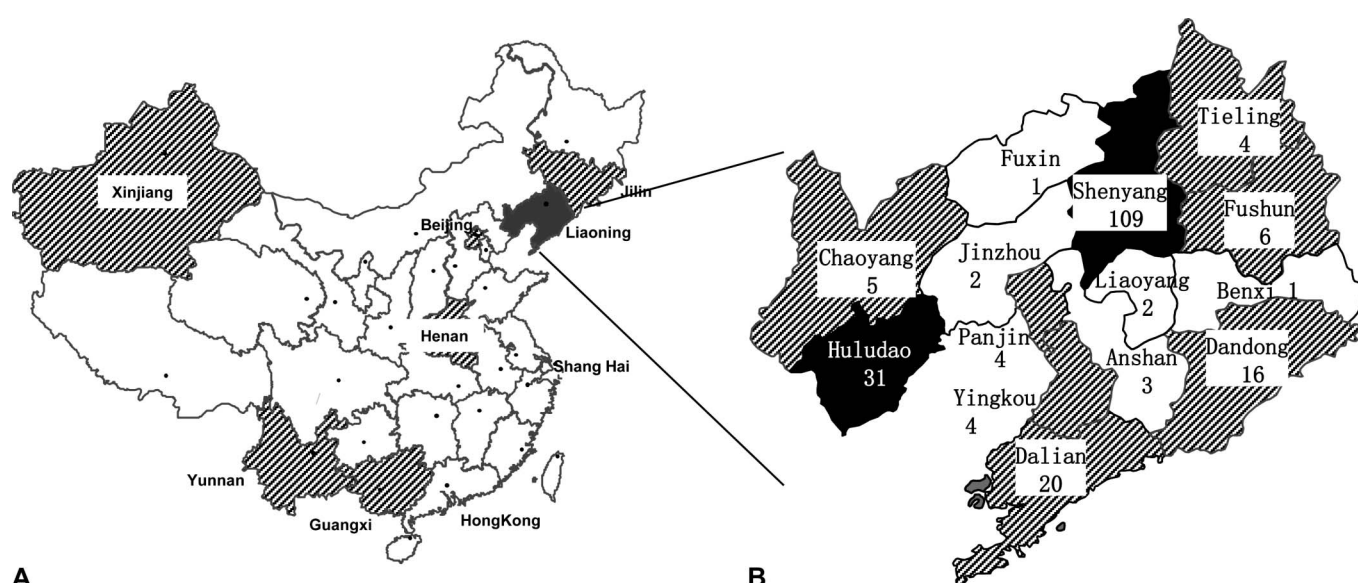


FIGURE 1. Map of the People's Republic of China (A) and Liaoning province (B), and the geographic distribution of studied HIV-1–infected cases in Liaoning.

controls to detect possible contamination during the procedure. To check for potential contamination, before analysis, the sequences obtained were compared with all known sequences in the HIV database by a Basic Local Alignment Search Tool (BLAST) search (<http://hiv-web.lanl.gov/content/index>).

Nucleotide Sequence Accession Numbers

The Genbank accession numbers of nucleotide sequences reported in this article are EF122502-EF122594, AY548279, and FJ531381-FJ531481.

RESULTS

The HIV-1/AIDS Epidemic in Liaoning

Liaoning province is located in the northeastern part of China (Fig. 1), the region traditionally referred to as Manchuria. Its geographic proximity to the Yellow Sea, the Bohai Gulf, and the Yalu River has made it an important base for industry and trade and it serves as the closest gateway to the Korean Peninsula. Administratively, Liaoning has a total of 14 counties and cities, of which the capital Shenyang is the largest (Fig. 1). Liaoning has an estimated population of 42 million, with Han as the ethnic majority (84%), followed by Manchu (13%), and Mongol (2%). The remaining 1% consists of Hui, Korean, and Xibo minorities.

Since the first HIV case was identified in Liaoning in 1991, a total of 990 HIV/AIDS cases had been reported by 2007. Although the cumulative number of HIV/AIDS cases remains relatively small, the rate of new case identification has increased rapidly in recent years. Although only 77 cases were found in 2004, 159 cases were identified in 2005, 216 cases in 2006, and 305 cases in 2007 (Fig. 2). This translates to an average annual rate of increase of approximately 60% over the last 3 years. Based on these reported cases, it is estimated that the total number of infected cases in Liaoning is about 5239, with province-wide prevalence rate of 0.0125%.

The recent increase in reported cases is also characterized by dramatic changes in risk factor distribution among infected individuals. Most notably, the proportion of cases transmitted through homosexual contact increased to 22%

between 2005 and 2007 compared with less than 1% in the previous years (Fig. 2, inset). On the other hand, the proportion of cases transmitted through blood donation and transfusion decreased from 34% to 8% during the same period. Other risk factors such as injection drug use (IDU), mother to child transmission, and heterosexual contact have contributed relatively stable proportions to the total. Of note, individuals with unknown risk factors have contributed a substantial proportion of the total throughout the years (Fig. 2, inset).

The rapid increase in MSM-related infections is a significant feature of the changing epidemiologic profile of HIV-1 in Liaoning. Although the HIV-1 prevalence rates among the IDU, female sex worker, and STD (out patients attending sexually transmitted disease clinics) groups remain relatively low and stable (below 0.5%) between 1999 and 2007, there is a 6-fold increase for MSM in 2006 and 2007, according to data from the Disease Control and Prevention center of Liaoning Province. Furthermore, more than 65% of infected cases are male. On average, >80% of infected individuals have been aged 20–40 years.

Although such dramatic increases in reported HIV/AIDS cases are primarily due to the active testing initiative launched by national and provincial health authorities in 2004, these could also indicate an expansion of the epidemic in Liaoning. Indeed, differentiating new infections from newly identified chronic infections will require more sophisticated techniques.

Different Risk Groups Dominated by Distinct Subtypes of HIV-1

To study the pattern of distribution of HIV-1 genotypes, we obtained 208 nucleotide sequences encompassing the *gag* RT gene. A phylogenetic approach was used to study the relationships among HIV-1 strains within Liaoning and with those from outside. The most obvious and consistent result from phylogenetic analysis was that there are 3 major subtypes and/or CRFs of HIV-1 currently circulating in Liaoning. They are those clustering closely with reference sequences CRF01_AE (01 AE.TH.90.CM240) or CRF15_01B (15 01B.TH.99.MU2079); those with subtype B (B.FR.83HXB2-LAI-IIIBBRU) or B' (B.CN.RL42); and those with subtype C

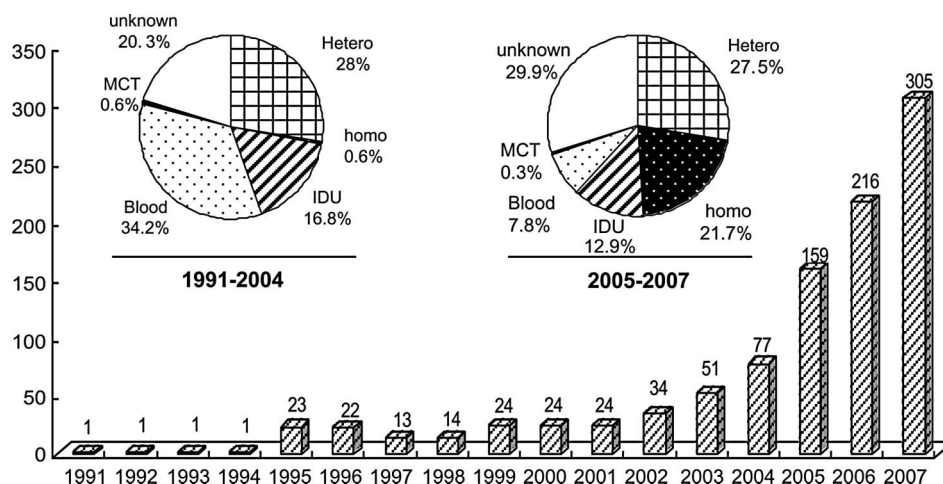


FIGURE 2. Accumulative reported HIV-1-infected cases in Liaoning between 1991 and 2007.

(C.IN.95.21068) or reference sequences CRF07_BC (07 BC.CN.97.CN54) or CRF08_BC (08 BC.CN.97.GX.6F). Approximately 43.3% of collected sequences were found in the CRF01_AE/CRF15_01B group, 41.3% in the subtype B/B' group, and 9.6% in the C/CRF07_BC/CRF08_BC group. The remaining 5.8% belong to other subtypes such as A1 or A2 or unknown recombinants. Of note, about 90.0% of infected IDUs had C/CRF07_BC/CRF08_BC viruses, whereas 92.3% of infections acquired through blood donation had subtype B or B', and 81.3% of MSM had CRF01_AE (Fig. 3A). The heterosexual contact risk group showed a variety of subtypes, with 37.1% subtype B or B', 12.9% CRF07_BC or CRF08_BC, 37.1% CRF01_AE, and the remaining 12.9% other subtypes (Fig. 3A). Furthermore, more than 96% of CRF01_AE infections were acquired through sexual transmission (67.8% MSM, 28.9% heterosexual risk) (Fig. 3B). Subtypes B and B', on the other hand, are strongly associated with blood donation (55.8%) and heterosexual contact (30.2%) and only weakly so with MSM (14.0%). Lastly, CRF07_BC and CRF08_BC are dominant in both the heterosexual contact (45.0%) and IDU (45.0%) risk groups (Fig. 3B).

Close Genetic Relationship Between HIV-1 Found in Liaoning and Those in High Prevalence Provinces and Regions

We have generated independent phylogenetic trees for the 3 major Liaoning HIV subtypes and CRFs; each of which is indicated by symbols corresponding to their known risk factor: homosexual (open triangle), heterosexual (closed triangle), IDU (closed square), and PBD/BT (closed circle). For clarity, only a few representative sequences from each risk group are shown, together with reference sequences from different HIV-1 subtypes (<http://hiv-web.lanl.gov/content/index>). As shown in Fig. 4, sequences identified in Liaoning have a close genetic relationship with those previously found in Henan, Guangxi, and Yunnan provinces. For instance, sequences in the AE group are interdigitized with those identified in Guangxi (01 AE.CN.97.GX2F) and in neighboring Thailand (01 AE.TH.90.CM240). Although those found in heterosexual contact (closed triangle) are spread along many branches, most MSM (open triangle) sequences stem from a single branch, suggesting a close relationship among MSM

in Liaoning (Fig. 4). This group contains sequences from both MSM and heterosexual risk individuals. The genetic distance within this group is as low as 0.027, suggesting the MSM and heterosexual contact groups are tightly linked. This finding perhaps relates to the fact that most of the MSM in China are bisexual and married with family.

Sequences in the subtype B/B' group are mostly clustered with B' sequences previously identified in China. Of the 2 groups of B' sequences, 1 group is related to the more recent sequences identified in Henan (B.CN.02.HNs4) and Hubei (B.CN.05.HB dw111), whereas the other group is related to early epidemic sequences found in Yunnan (B.CN.x.RL42). As in the AE group, individuals infected through heterosexual contact have sequences scattered along many branches in both B and B' groups, whereas those from FBD are derived from a single branch, suggesting the tight relationship among blood donors in Liaoning and their genetic relatedness with those from those high prevalence provinces. In addition, sequences from blood donors are tightly grouped with those from heterosexual risk individuals (Fig. 4), and their genetic distance is as low as 0.038. This result may indicate that HIV-1 has begun to spread from the traditionally high-risk blood donor population to some in the general public through heterosexual contact. Lastly, some MSM were also found to have subtype B sequences similar to typical North American strains such as B.FR.83.HXB2-LAI-IIIB-BRU.

Sequences in the BC groups fall into 2 previously identified recombinant forms, CRF07_BC and CRF08_BC (Fig. 4). They are predominately comprised of IDUs (closed square) and heterosexual risk individuals (closed triangle) and are genetically indistinguishable from those previously identified in Yunnan (07 BC.CN.97.CN54), Guangxi (08 BC.CN.97.GX.6F), and India (C.IN.95.21068) (Fig. 4). One sequence loosely groups with subtype C, and its true subtype assignment needs to be further verified.

Recombinant Forms of HIV-1 Found in Liaoning Similar to Those Found in High Prevalence Provinces and Regions

We analyzed 62 HIV-1 sequences from Liaoning spanning the entire 2.6 kilobase *gag* RT gene (nucleotide

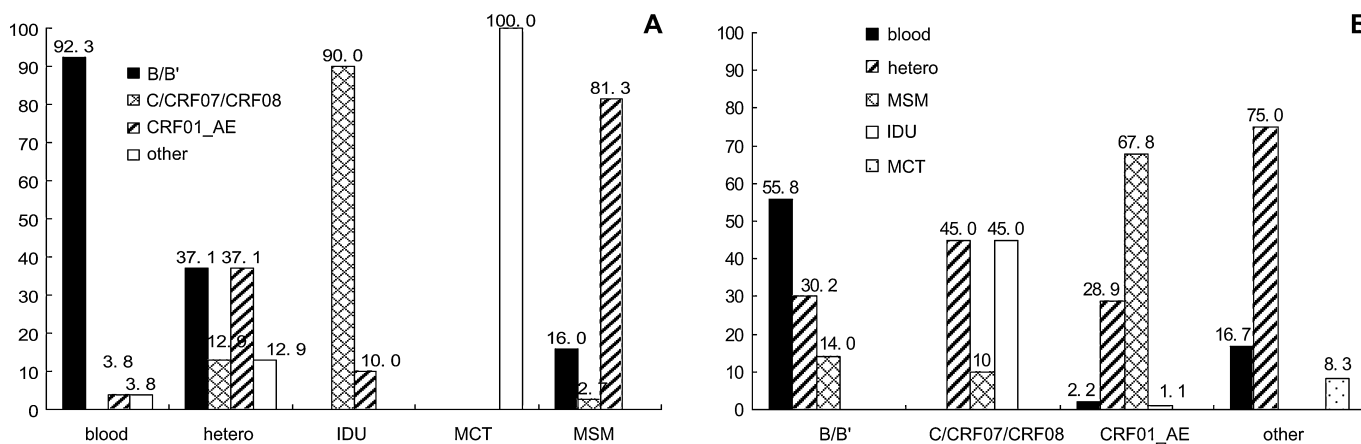


FIGURE 3. Correlation between risk factors and HIV-1 subtypes and CRF. A, Percentage of viral subtypes and CRF in each of risk group studied; and B, Percentage of risk groups in each of the viral subtype and CRF.

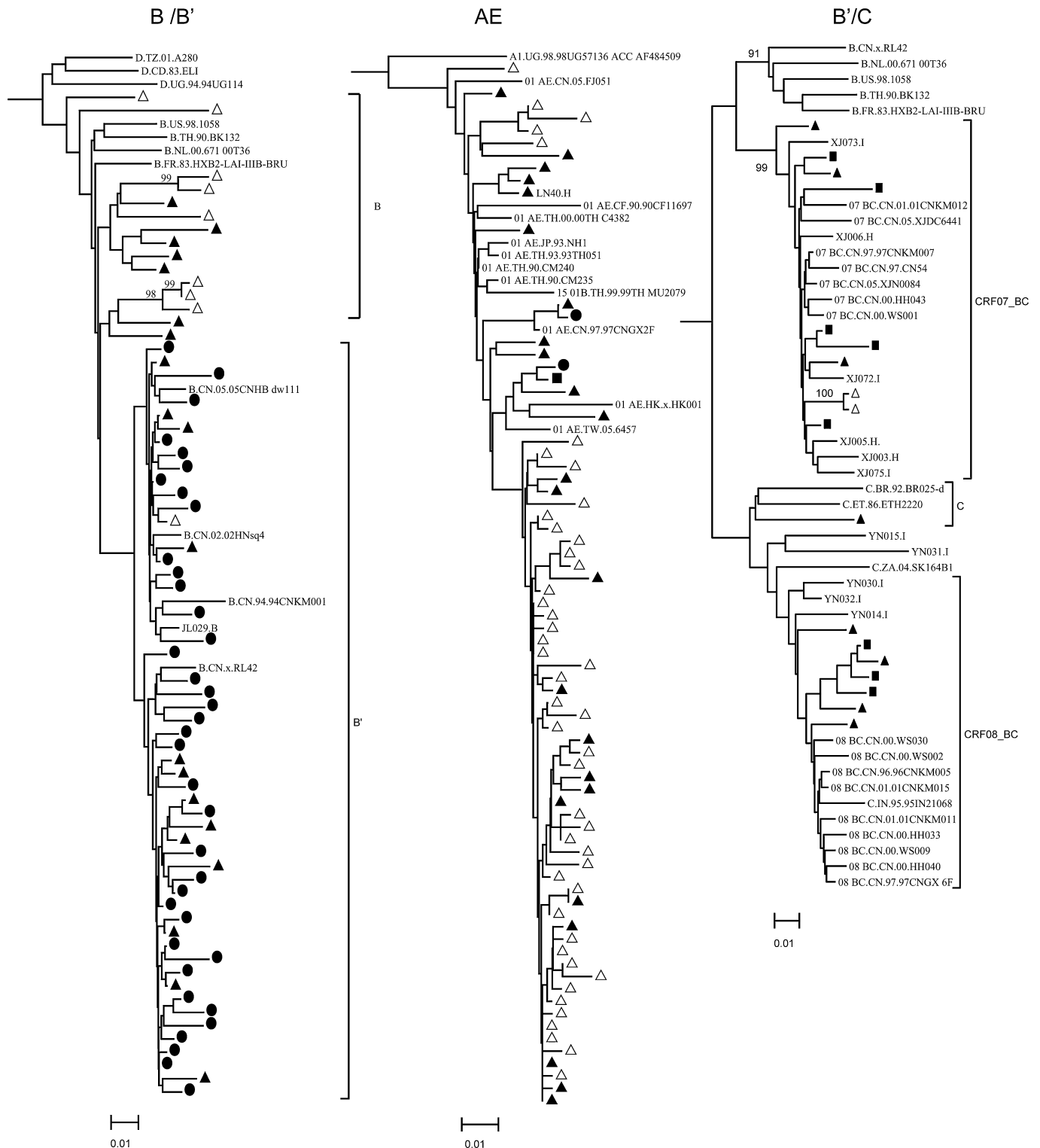


FIGURE 4. Neighbor-joining trees showing the 3 major circulating HIV-1 subtypes and CRF in Liaoning and their genetic relationship with those identified in high prevalence provinces. Individual sequences are indicated by symbols corresponding to their known risk factor: homosexual (open triangle), heterosexual (closed triangle), IDU (closed square), and FBD (closed circle). For clarity only a few representative sequences from each risk groups are shown, together with reference sequences from different HIV-1 subtypes (<http://hiv-web.lanl.gov/content/index>). The bootstrap values of 1000 replicates are labeled on the major branches.

position relative to HXB2 genome: 790–3421). Consistent with phylogenetic analysis, all CRF08_BC or CRF07_BC viruses had typical recombination features similar to those previously reported from Guangxi and Yunnan (Fig. 5). None of the CRF01_AE sequences showed any intersubtype recombination in the fragment studied. However, 1 sequence was found that is likely a new subtype CRF01_AE/B recombinant (Fig. 5). Further studies will be needed to characterize this strain in the context of full viral genome.

DISCUSSION

In this report, we have characterized the epidemiologic and genetic features of HIV-1 infection in relatively low prevalence Liaoning province. We believe this is the first time that comparative studies have been conducted between HIV-1 in high and low prevalence provinces in China. We have shown that in recent years, although HIV-1 prevalence in Liaoning has remained low, the rate of new infection has increased rapidly. The most astonishing finding is a dramatic increase in HIV-1 infection among MSM, accounting for more than 26% of the entire infected populations in 2007. In addition, the component of infections through heterosexual contact has reached 28%. In other words, sexual contact has become the dominant mode of HIV-1 transmission (26% + 28% = 54%) among infected cases.

Furthermore, major subtypes and CRFs of HIV-1 previously identified in Henan, Guangxi, and Yunnan have all been identified in Liaoning. Although there are some reports trying to illustrate the temporal and spatial relationship

between HIV-1 in high and low prevalence provinces and regions,^{17,18} it is impossible to describe exactly the chain of transmission events that led to the current epidemiologic picture in Liaoning. Nevertheless, our results provide some evidence that the HIV-1/AIDS epidemic in Liaoning is the result of multiple introductions of various HIV-1 strains from high prevalence provinces and regions. Liaoning, therefore, serves as the most critical base for the introduction and subsequent spread of HIV-1 in northeastern China.

As in high prevalence regions, there are close relationships between HIV-1 subtypes/CRFs and certain risk behaviors in Liaoning. For example, subtype B' and B sequences are predominantly found in paid blood donors, CRF07_BC and CRF08_BC mainly in IDU, and CRF01_AE in MSM, indicating that these viruses were introduced to Liaoning through their respective risk behaviors. However, such correlation has been compromised to a large extent in Liaoning compared with that in the original high prevalence provinces. Notably, multiple subtypes and CRFs have been identified within the heterosexual risk group, suggesting that HIV-1 strains from historically high-risk groups have been transmitted to the general population via heterosexual contact. Furthermore, the dramatic expansion of HIV-1 infection among MSM is comprised largely of CRF01_AE sequences. The strong genetic relatedness among CRF01_AE viruses demonstrates the rapid and perhaps recent dissemination of viruses within the MSM group in Liaoning.¹⁹ This hypothesis is consistent with epidemiologic data showing 2006 as the turning point when there was a significant increase in HIV-1 prevalence among MSM. Lastly, as multiple subtypes and

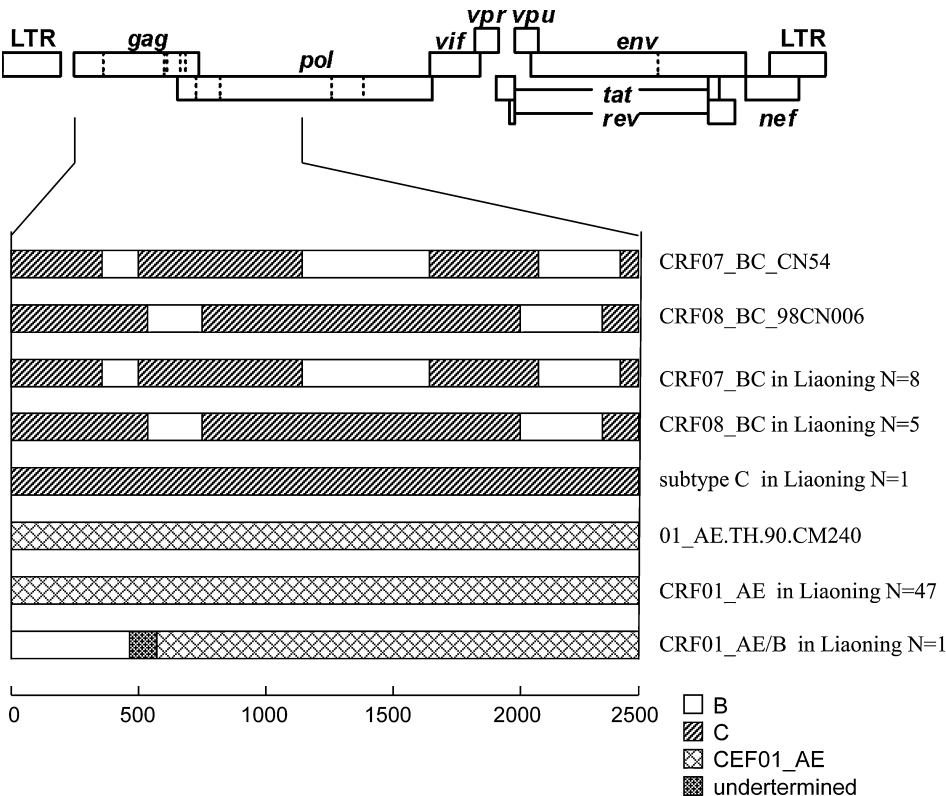


FIGURE 5. Bootscanning Analysis of HIV-1 gag-reverse transcriptase Sequences of 62 study participants. Participants included 8 CRF07_BC virus, 5 CRF08_BC virus, 1 pure subtype C viruses 47 CRF01_AE, and 1 new CRF01_AE and B recombinant virus 02CNRL42. The bootstrap values are based on 500 replicates using the neighbor-joining method, with window: 200 basepairs, step 20 basepairs, Gapstrip:on, Kimura(2 parameter), T/t:2.0.

CRFs are circulating within the same population in geographic proximity, it is expected that further recombination events are likely to occur and generate novel recombinants. Indeed, in this instance, we have identified a new recombinant between CRF01_AE and B', although this must be further verified.

In summary, we have for the first time conducted province-wide epidemiologic and genetic studies on HIV-1 in low prevalence Liaoning province. Our data strongly indicate the close relationship between HIV-1 sequences in Liaoning and those in high prevalence regions. HIV-1 has therefore been able to overcome our preventive measures and continues to spread from the high prevalence southwest to the low prevalence northeast region of China. As the epidemic continues to spread, it is expected that viral genetic diversity will increase, which will inevitably pose more challenges to vaccine, drug, and microbicide development. It is high time to implement more effective and integrated measures to reduce the further dissemination of these viruses—in low prevalence areas like Liaoning and throughout the world's most populous nation.

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